

DRONES, UAVS AND ROBOTICS POLICY

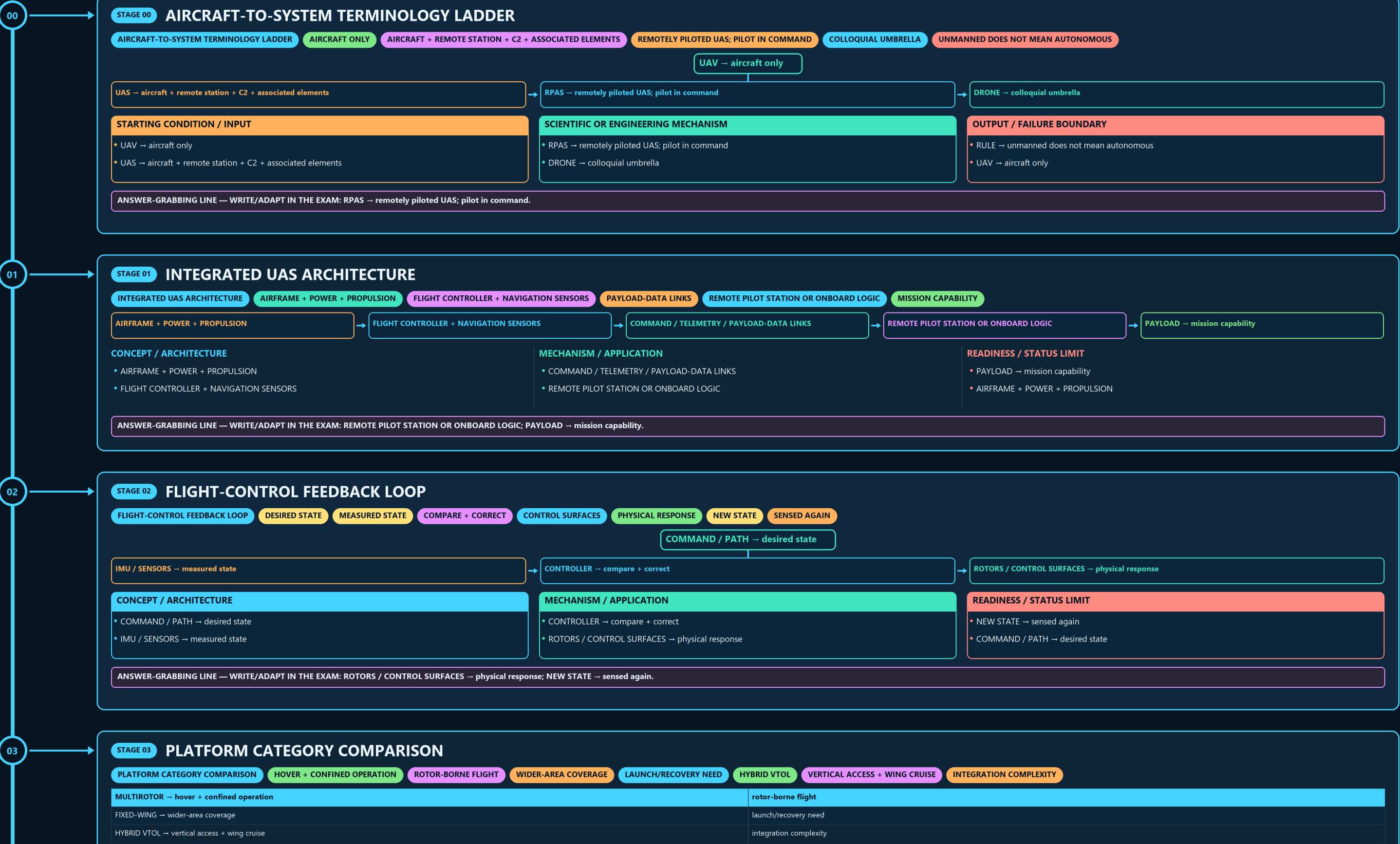
AIRCRAFT-TO-SYSTEM TERMINOLOGY LADDER → INTEGRATED UAS ARCHITECTURE → FLIGHT-CONTROL FEEDBACK LOOP → PLATFORM CATEGORY COMPARISON → HUMAN-CONTROL CONTINUUM → SWARM COORDINATION MESH

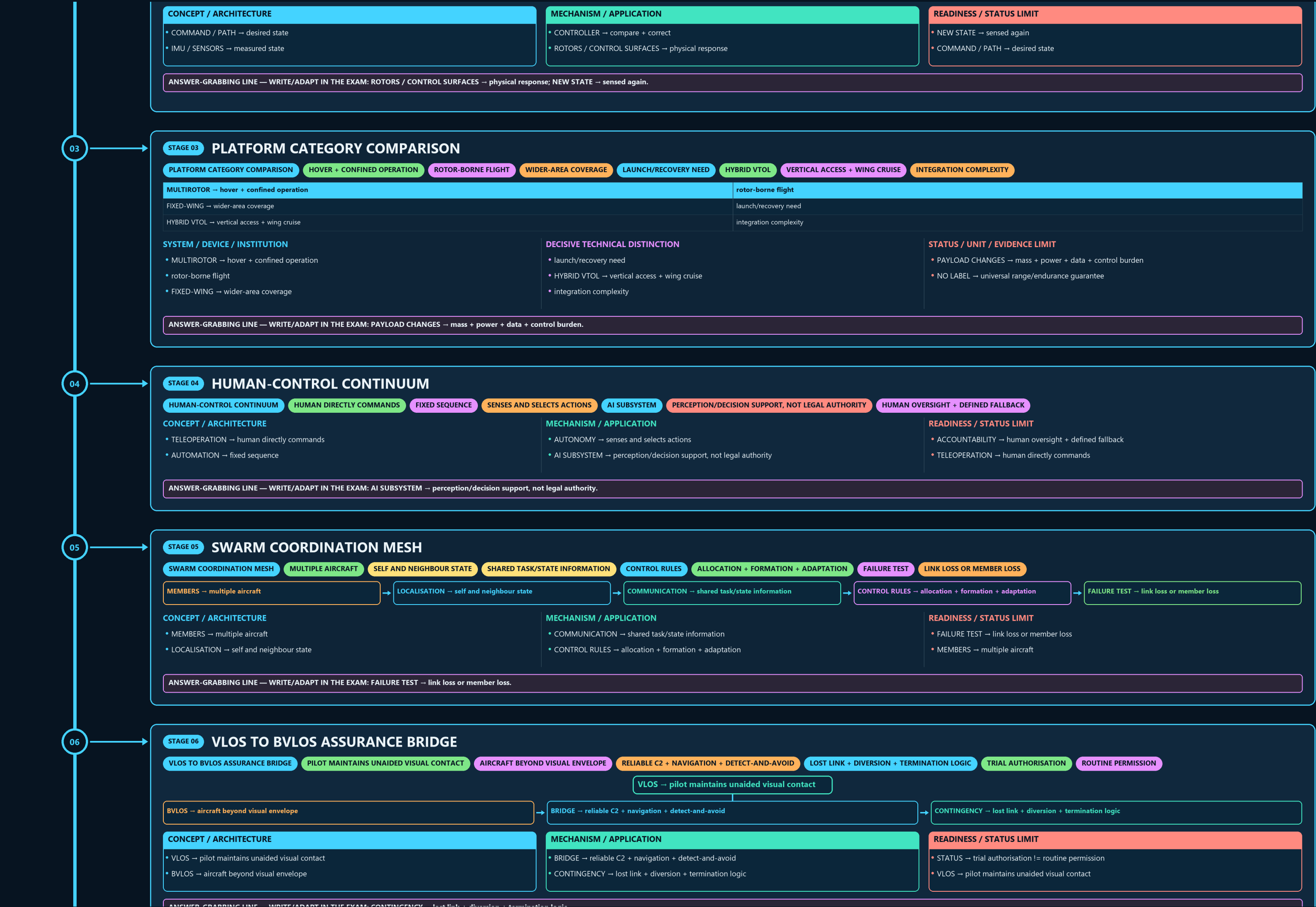
Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g3 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles





06

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FAILURE TEST → link loss or member loss.

STAGE 06

VLOS TO BVLOS ASSURANCE BRIDGE

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PILOT MAINTAINS UNAIDED VISUAL CONTACT

AIRCRAFT BEYOND VISUAL ENVELOPE

RELIABLE C2 + NAVIGATION + DETECT-AND-AVOID

LOST LINK + DIVERSION + TERMINATION LOGIC

TRIAL AUTHORISATION

ROUTINE PERMISSION

VLOS → pilot maintains unaided visual contact

BVLOS → aircraft beyond visual envelope

BRIDGE → reliable C2 + navigation + detect-and-avoid

CONTINGENCY → lost link + diversion + termination logic

CONCEPT / ARCHITECTURE

- VLOS → pilot maintains unaided visual contact
- BVLOS → aircraft beyond visual envelope

MECHANISM / APPLICATION

- BRIDGE → reliable C2 + navigation + detect-and-avoid
- CONTINGENCY → lost link + diversion + termination logic

READINESS / STATUS LIMIT

- STATUS → trial authorisation != routine permission
- VLOS → pilot maintains unaided visual contact

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CONTINGENCY → lost link + diversion + termination logic.

07

STAGE 07

COUNTER-UAS DECISION FUNNEL

COUNTER-UAS DECISION FUNNEL

CLASSIFY AND MAINTAIN PICTURE

AUTHORITY + PROPORTIONALITY + SURROUNDINGS

OTHER AUTHORISED RESPONSE

JAMMING/SPOOFING CAN AFFECT LEGITIMATE SYSTEMS

CONCEPT / ARCHITECTURE

- DETECT → radar
- IDENTIFY / TRACK → classify and maintain picture

MECHANISM / APPLICATION

- DECIDE → authority + proportionality + surroundings
- MITIGATE → electronic

READINESS / STATUS LIMIT

- other authorised response
- LIMIT → jamming/spoofing can affect legitimate systems

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MITIGATE → electronic | capture | other authorised response.

08

STAGE 08

CIVILIAN APPLICATION EVIDENCE GRID

CIVILIAN APPLICATION EVIDENCE GRID

WILDLIFE OBSERVATION

DISASTER ASSESSMENT

PILOT OR AUTHORISED MISSION, NOT BLANKET SERVICE

IMAGERY + INTERPRETATION + FIELD VERIFICATION

CONCEPT / ARCHITECTURE

AGRICULTURE → monitor / map / spray

SCIENCE → volcano / wildlife observation

MECHANISM / APPLICATION

GOVERNANCE → survey / inspection / disaster assessment

LOGISTICS → pilot or authorised mission, not blanket service

READINESS / STATUS LIMIT

EVIDENCE → imagery + interpretation + field verification

AGRICULTURE → monitor / map / spray

CONCEPT / ARCHITECTURE

- AGRICULTURE → monitor / map / spray
- SCIENCE → volcano / wildlife observation

MECHANISM / APPLICATION

- GOVERNANCE → survey / inspection / disaster assessment
- LOGISTICS → pilot or authorised mission, not blanket service

READINESS / STATUS LIMIT

- EVIDENCE → imagery + interpretation + field verification
- AGRICULTURE → monitor / map / spray

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE → imagery + interpretation + field verification.

09

STAGE 09

SAFETY PRIVACY AND LIABILITY SHIELD

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AIRWORTHINESS + MAINTENANCE + FAILSAFES

PILOT + WEATHER + ROUTE + PAYLOAD DISCIPLINE

PURPOSE + ACCESS + RETENTION + GEOSPATIAL SENSITIVITY

C2/NAVIGATION/PAYLOAD-LINK RESILIENCE

TRACE OPERATOR/SOFTWARE/MANUFACTURER FUNCTION

MUST REMEMBER

DRONES COMBINE AIRFRAME, PROPULSION, CONTROL, COMMUNICATION/NAVIGATION

ENGINEERING → airworthiness + maintenance + failsafes

OPERATIONS → pilot + weather + route + payload discipline

DATA → purpose + access + retention + geospatial sensitivity

CYBER → C2/navigation/payload-link resilience

LIABILITY → trace operator/software/manufacture function

CONCEPT / ARCHITECTURE

- ENGINEERING → airworthiness + maintenance + failsafes
- OPERATIONS → pilot + weather + route + payload discipline

MECHANISM / APPLICATION

- DATA → purpose + access + retention + geospatial sensitivity
- CYBER → C2/navigation/payload-link resilience

READINESS / STATUS LIMIT

- LIABILITY → trace operator/software/manufacture function
- MUST REMEMBER: Drones combine airframe, propulsion, control, communication/navigation,...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MUST REMEMBER: Drones combine airframe, propulsion, control, communication/navigation,.

10

STAGE 10

INDIAN GOVERNANCE RESPONSIBILITY MAP

DRONES, UAVS AND ROBOTICS POLICY • TILE 3/4 • same master rows 4816-8050 of 10458 • continues below

09

STAGE 09

SAFETY PRIVACY AND LIABILITY SHIELD

AIRWORTHINESS + MAINTENANCE + FAILSAFESPILOT + WEATHER + ROUTE + PAYLOAD DISCIPLINEPURPOSE + ACCESS + RETENTION + GEOSPATIAL SENSITIVITYC2/NAVIGATION/PAYLOAD-LINK RESILIENCETRACE OPERATOR/SOFTWARE/MANUFACTURER FUNCTION

MUST REMEMBERDRONES COMBINE AIRFRAME, PROPULSION, CONTROL, COMMUNICATION/NAVIGATION

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CONCEPT / ARCHITECTURE

ENGINEERING → airworthiness + maintenance + failsafes
OPERATIONS → pilot + weather + route + payload discipline

MECHANISM / APPLICATION

DATA → purpose + access + retention + geospatial sensitivity
CYBER → C2/navigation/payload-link resilience

READINESS / STATUS LIMIT

LIABILITY → trace operator/software/manufacture function
MUST REMEMBER: Drones combine airframe, propulsion, control, communication/navigation,...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MUST REMEMBER: Drones combine airframe, propulsion, control, communication/navigation,.

10

STAGE 10

INDIAN GOVERNANCE RESPONSIBILITY MAP

CIVIL REGULATIONAIRSPACE AND TRAFFIC SERVICESAVIATION SECURITYDIGITAL SKYWORKFLOW PLATFORM, NOT ATCCLOSE DISTINCTIONDRONE IS NOT NECESSARILY AUTONOMOUS, REMOTE PILOTING IS NOT

DGCA → civil regulation / certification

AAI / ATC → airspace and traffic services

BCAS → aviation security

MOCA → policy / notification

CONCEPT / ARCHITECTURE

MOCA → policy / notification
DGCA → civil regulation / certification

MECHANISM / APPLICATION

AAI / ATC → airspace and traffic services
BCAS → aviation security

READINESS / STATUS LIMIT

DIGITAL SKY → workflow platform, not ATC
CLOSE DISTINCTION: Drone is not necessarily autonomous, remote piloting is not AI,...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Drone is not necessarily autonomous, remote piloting is not AI,.

11

STAGE 11

APPROVAL-TO-OUTCOME STATUS RAIL

LEGAL TEXTTYPE CERTIFICATEDISTINCT COMPLIANCE CONDITIONSAUTHORISED OR SUPPORTED ROUTENOT OPERATIONAL DEPLOYMENTNOT VERIFIED SAFETY, SCALE OR OUTCOME EVIDENCE LIMIT

RULE / AMENDMENT → legal text

TYPE CERTIFICATE / UIN / RPC → distinct compliance conditions

TRIAL / SCHEME / INCENTIVE → authorised or supported route

PROCUREMENT / DELIVERY → not operational deployment

DEPLOYMENT → not verified safety, scale or outcome

CONCEPT / ARCHITECTURE

RULE / AMENDMENT → legal text
TYPE CERTIFICATE / UIN / RPC → distinct compliance conditions

MECHANISM / APPLICATION

TRIAL / SCHEME / INCENTIVE → authorised or supported route
PROCUREMENT / DELIVERY → not operational deployment

READINESS / STATUS LIMIT

DEPLOYMENT → not verified safety, scale or outcome
EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: State platform class, weight/category,...

MECHANISM / VERDICT — EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: State platform class, weight/category,...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: State platform class, weight/category,.

E

EXTRA

SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENTTHE MOST IMPORTANT ANALYTICAL SPLIT IS BETWEEN AIRFRAME MANUFACTURINGTHE LATTER DEPENDS ON SENSORS, SOFTWARE, COMMUNICATIONS AND PAYLOADDRONE POLICY SITS AT THE INTERSECTION OF AVIATION LAW

ROBOTICS CAPABILITY IS INCREASINGLY DUAL-USEAGRICULTURE SPRAYING AND LOGISTICS CAN COEXIST WITH DEFENCE-SUPPORT ORRULE-MAKING AND SCHEME-LEVEL ANCHOR.COMPLIANCE AND CIVIL-DRONE REGULATORY AUTHORITY.

OPTIONAL SCIENTIFIC / ENGINEERING DEPTH

CAUTION: The most important analytical split is between airframe manufacturing and mission capability
the latter depends on sensors, software, communications and payload integration.
CAUTION: Drone policy sits at the intersection of aviation law, digital governance, privacy, internal security and industrial policy.
CAUTION: Robotics capability is increasingly dual-use: agriculture spraying and logistics can coexist with defence-support or surveillance relevance.

READINESS, UNIT OR STATUS LIMIT

MoCA: rule-making and scheme-level anchor.
DGCA: compliance and civil-drone regulatory authority.
Digital Sky: public-facing airspace and permission architecture.
PLI for drones and drone components: industrial push for indigenous ecosystem capability.

COMPARATIVE TECHNOLOGY / GOVERNANCE NUANCE

Training/certification ecosystem: necessary for safe scaling of operations.
Agriculture: precision application and monitoring can reduce time and improve coverage.
Disaster response: rapid imaging and access to difficult terrain improves first-response intelligence.
Surveying/logistics/infrastructure inspection: drones reduce time and increase reach.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.